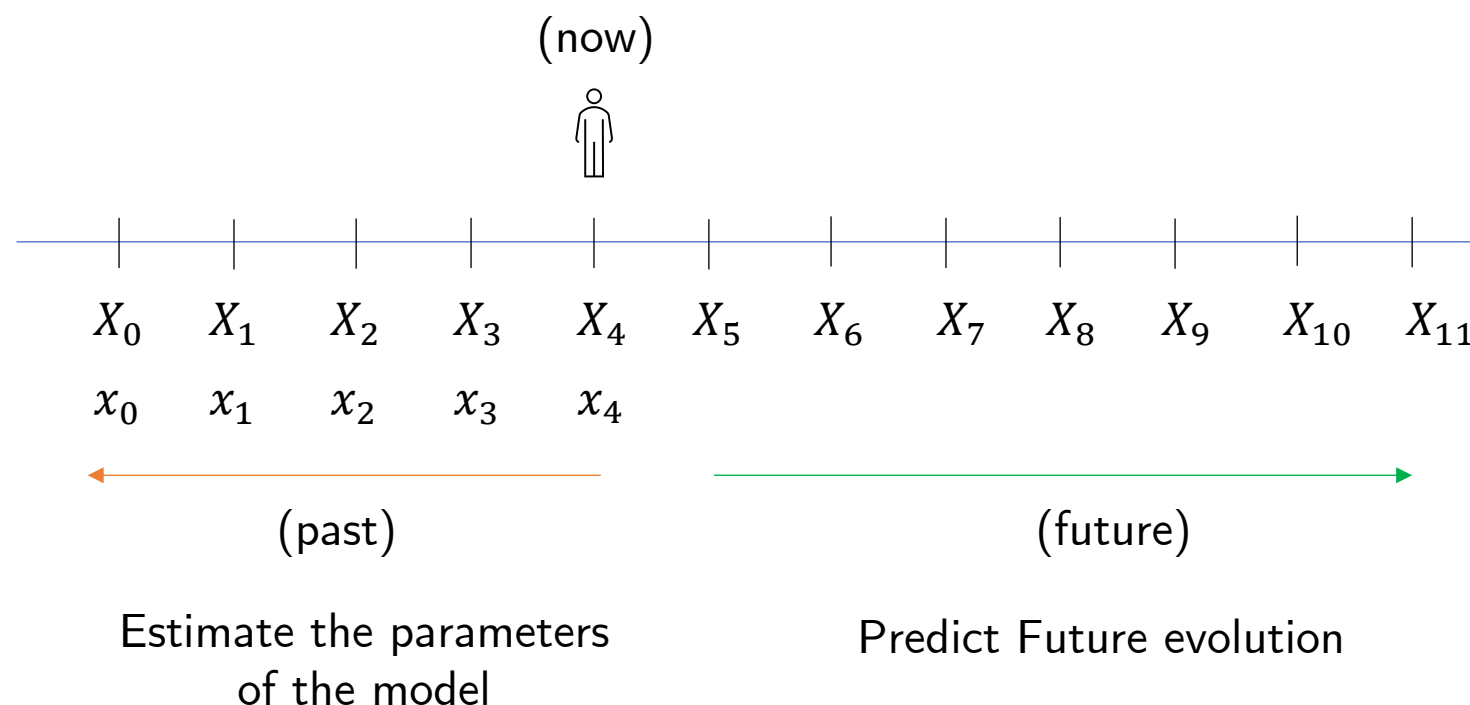


Stochastics

(1. Stochastic Processes)

Stochastic process

- ▶ A **stochastic process** is a set of ordered random variables indexed by time. X_t is a random variable that models the value of the stochastic process at time t .
- ▶ For example, X_t may represent stock price at time t . At any time, we will have a past and future of the stochastic process. The past is used to estimate model parameters. The model is then used to predict future (or future distribution)





Time domain and state space

► Stochastic process is a collection of random variables X_t , one for each time t in some set J

$$\{X_t : t \in J\}$$

The set J serves as the domain (**time domain**) of the process. The set of values the random variables X_t can take is called the **state space S** . The set of random variables maybe dependent on each other. So, to describe their statistical properties we need to know their **joint distribution** (for example, distribution of (X_9, X_{10}) in the previous example)

The random variables in the set may not be identically distributed as well. But identically distributed random variables are important in stochastic processes. If the random variables are both independent and identically distributed, they are referred to as **i.i.d**

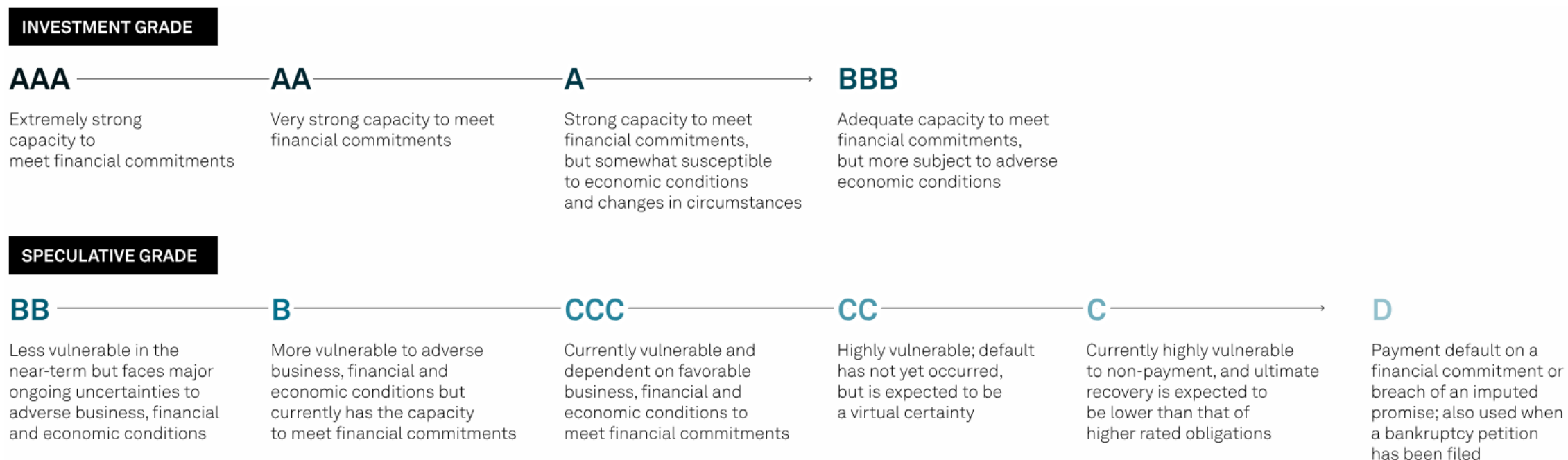
Question : Let's say we model the S&P500 daily closing price each day by the process $\{X_t\}$. Are the state space and time domain for the process discrete or continuous ?

Choice of discrete vs continuous

- ▶ Stock prices change frequently and by minimum tick size. So, according to reality the state space or time domain both may be discrete. However, managing so many different discrete states and associated probabilities might be difficult to model.
- ▶ Moreover, the frequency of recording the observations for the same stochastic process may be different due to physical limitation of the device or other constraints or use case dependent. For example, to model stock price for say 1 day Value-at-Risk calculation, daily closing price may be sufficient. However, for HFT, we need data points every minute.
- ▶ In stochastic modeling for financial variables, it's common to use continuous model when the discrete levels become enormous (example stock price, interest rates, FX rates etc.). This allows us to use the entire machinery of calculus (although a different type of calculus!)

Discrete time Discrete space (DTDS)

- ▶ A Rating agency (e.g., S&P Global) reviews the credit worthiness of companies every quarter. The allowed ratings are given as $\{ \text{AAA, AA, A, BBB, BB, B, CCC, CC, D} \}$. The state space is therefore discrete. The time domain is discrete as well as the ratings are updated once a quarter.



Continuous Time Discrete Space (CTDS)



- ▶ The state of an obligor (default, no default) is an example of a discrete state space (only two states) and continuous time domain as the customer can default any time.
- ▶ An insurance company classifies policyholder as {Healthy, Sick, Dead}. The transition of a policyholder from one state to another can happen at any moment. This is another example of discrete state space and continuous time domain.

(State Space) : $\{D, ND\}$



(State Space) : $\{H, S, D\}$



Discrete Time Continuous Space (DTCS)



- ▶ Time Series processes such as Autoregressive are examples of discrete time continuous space. Example would be daily closing stock price or quarterly GDP growth rates, unemployment rates etc.

☆ Unemployment Rate - 20 Yrs. & over (LNS14000024)

Observations ▾

Apr 2025: **3.8**

Updated: May 2, 2025 7:48 AM CDT

Next Release Date: Jun 6, 2025

Units:

Percent,
Seasonally Adjusted

Frequency:

Monthly

1Y

5Y

10Y

Max

Edit Graph

1948-01-01

to

2025-04-01

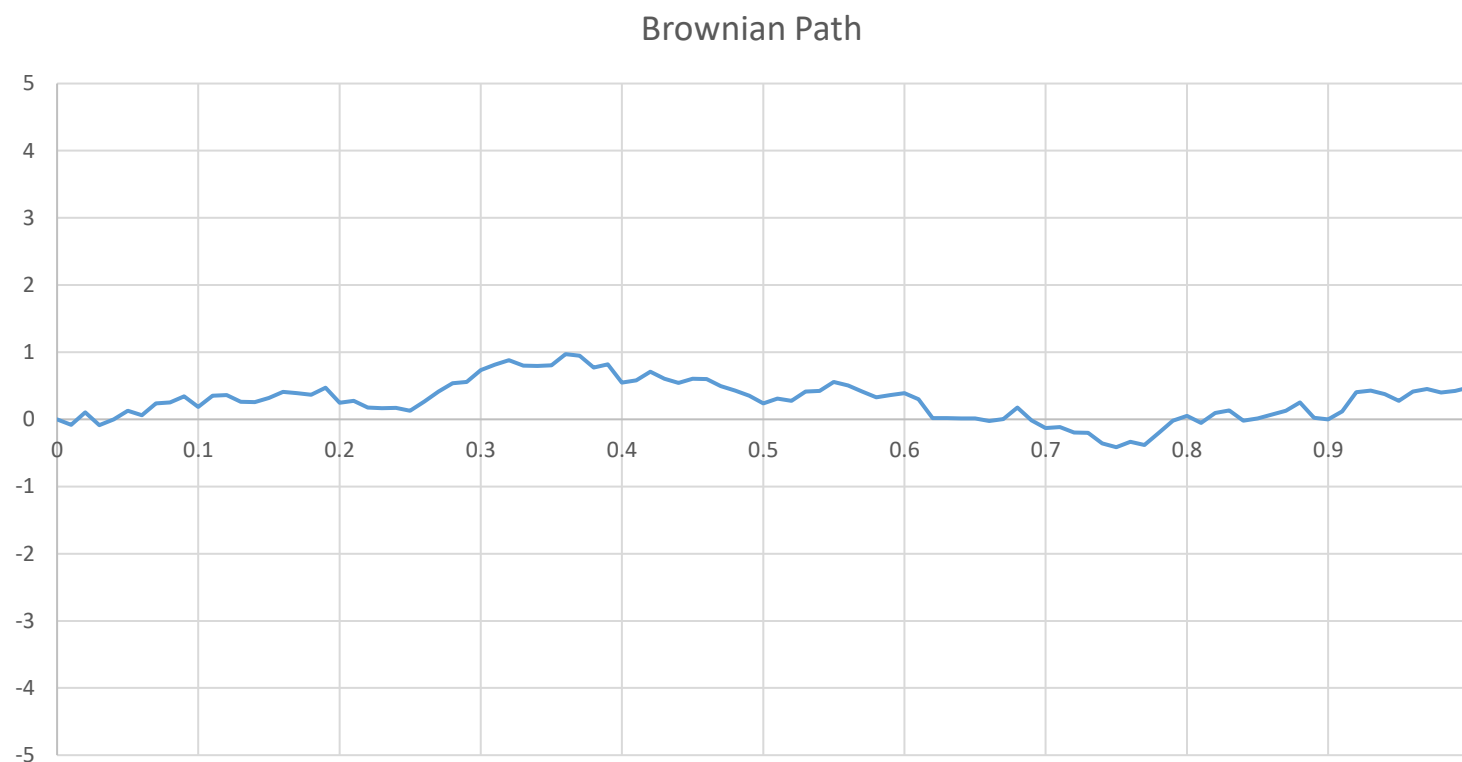
Download



Continuous Time Continuous Space (CTCS)



- ▶ Continuous time and continuous space processes provide the ultimate flexibility as it allows us to use calculus. Examples of such processes are **Brownian Motion** (Wiener Process) which is the primary component of stochastic calculus.



Question



You are thinking of moving to live in Edinburgh. As part of your research into what it's like to live in Edinburgh you have collected the following sets of data:

- the maximum daily temperature each day since 1 January 2006
 - whether or not it rained for each day since 1 January 2006
 - the number of cyclists injured in road accidents since 1 January 2006.
- (i) For each data set choose an appropriate state space and a time domain. In each case state whether the state space and the time domain are discrete or continuous, and give the units of measurement, e.g., dollars, weeks.
- (ii) Imagine that you have data for each process. Draw sketches to display these samples of data.

Mixed Type



Some stochastic processes can be a mixture of two separate stochastic processes. For example, consider a car business. Each car sells for either £40,000 or £50,000 (cash only) and you sell about 10 to 20 each year. The “life blood” of the business is the regular servicing and maintenance of the cars you have sold previously.

The revenue balance is a mixture of two stochastic processes.

- 1) A lot of regular small transactions arising from maintenance and servicing is a continuous state and continuous time process.
- 2) High value discrete sale revenues from car sales is a discrete state and continuous time process.

There is a specific model we use to describe stock price process where we use both diffusion (Brownian Motion) and Poisson Jumps to model the stock price. This is another example of mixed type. (More later!)

Model Specification



Having defined the state space and time domain, we now need to define the process itself. Now to define the process (a collection of random variables), we would ideally need to define the joint distribution of $(X_1, X_2, \dots, X_n)$. In practice, this will be a formidable task. Therefore, we often define an **auxiliary process** which indirectly completely defines the stochastic process.

As an example, consider a model where the stock price either increases or decreases by \$1 every day. The up-move and the down-move have probabilities p and $(1 - p)$ respectively. For simplicity, assume the changes are independent. Let the price of the stock price at time t is S_t with $S_0 = 100$. This explicitly pins down the entire stochastic process.

This model is called a **simple random walk** process with barrier at 0. The time domain (measured in days) is $J = \{0, 1, 2, \dots\}$. The state space is the set of non-negative integers $\{0, 1, 2, \dots\}$.

Here we have defined the process for the **increments** $X_t = S_t - S_{t-1}$ which is the daily change in the stock price.

The Simple Random Walk

▶ A simple random walk process $Y_t = Y_{t-1} + X_t$, where X_t is defined as follows.

$$X_t = \begin{cases} 1 & , \quad p \\ -1 & , \quad 1 - p \end{cases}$$

A **simple symmetric random walk** is the one where $p = 0.5$

We have assumed that the increments (X_t) are i.i.d. Each increment serves as an independent step. The series of steps constitute the walk. Most often increments have nicer properties than the process itself.

Question – Is the original stochastic process (Y_t) i.i.d if the increments (X_t) are i.i.d?

Question



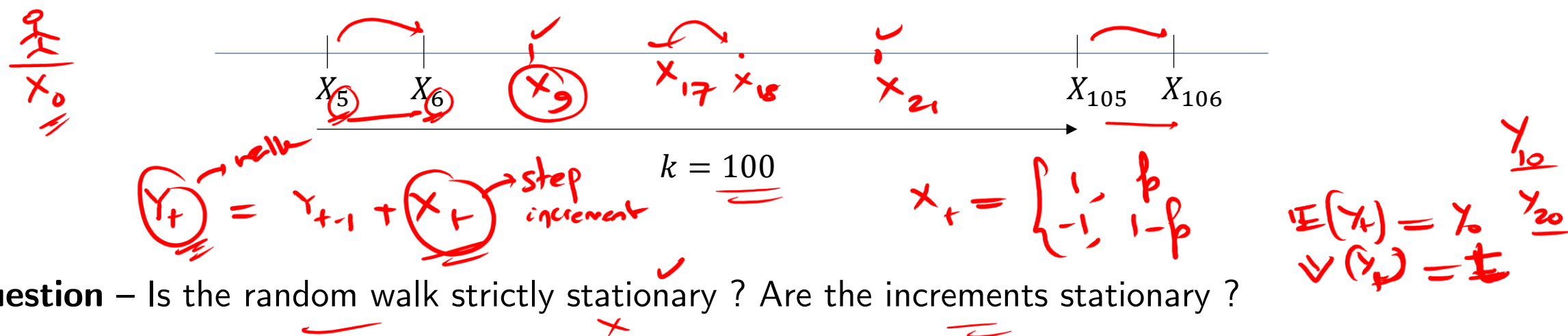
- 1) Find the Mean and Variance of the increment and the original process as a function of t if $p = 0.5$.
- 2) Calculate the following joint probabilities
 - $\mathbb{P}(Y_2 = 102, Y_5 = 103 \mid Y_0 = 100)$
 - $\mathbb{P}(Y_2 = 100, Y_4 = 103 \mid Y_0 = 100)$

Stationarity (I)

Stationarity means the distributional properties do not change over time. There are two flavors of stationarity

Strict Stationarity : The joint distribution of $(\overbrace{X_{t_1}, X_{t_2}, \dots, X_{t_n}})$ and $(\overbrace{X_{k+t_1}, X_{k+t_2}, \dots, X_{k+t_n}})$ are the same for all $(t_1, t_2, \dots, t_n)$ and $(k + t_1, k + t_2, \dots, k + t_n)$ in J . In other words, the joint distribution is invariant under time shifts

As an example, the distribution of (X_5, X_6) is same as the distribution of (X_{105}, X_{106}) .

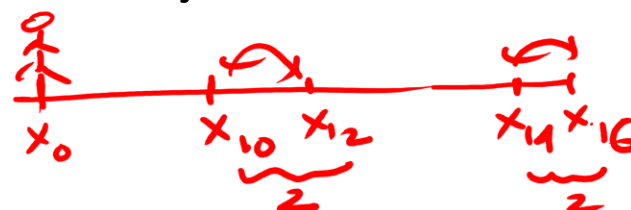


Question – Is the random walk strictly stationary? Are the increments stationary?

Stationarity (II)

Weak Stationarity : Strict stationarity is a stringent requirement which is not used in practice. We have a more practical flavor of stationarity which is heavily used in Time Series, it's called "**Weak Stationarity**". The requirements for Weak Stationarity are

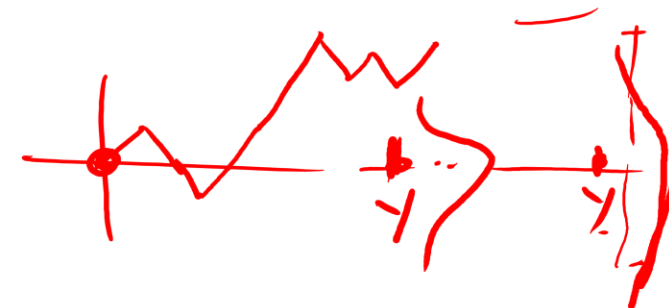
1. Mean of the process is constant
2. Covariance of $(X(s), X(t))$ only depend on the lag i.e., $(t - s)$



$$\begin{aligned} \text{Cov}(Y_s, Y_s) &\rightarrow \text{Var}(Y_s) \\ &= \text{Cov}(Y_t, Y_t) \rightarrow \text{Var}(Y_t) \end{aligned}$$

That's why this version of stationarity is also called as '**Covariance Stationarity**'. This also indirectly implies that variance of the process is constant i.e., when $s = t$. Entire timeseries infrastructure is based on this property of stationarity. If a process is strictly stationary, then it also trivially satisfies weak stationarity.

$$\begin{aligned} (Y_t) &\rightarrow \text{Var}(Y_s) = s \\ &\quad \text{Var}(Y_t) = t \end{aligned}$$



Covariance (Recap)

Quick Recap on some covariance examples

$$1. \text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$2. \text{Cov}(X, c) = 0$$

$$3. \text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$$

$$4. \text{Cov}(W, X + Y) = \text{Cov}(X, W) + \text{Cov}(Y, W)$$

$$(a+b)(c+d) \\ ac + ad + bc + bd$$

Question –

$$1) \text{ Show the following result } \text{Cov}(aX + bY, cW + d) = ac \text{Cov}(W, X) + cb \text{Cov}(W, Y)$$

$$2) \text{ Show that for simple symmetric random walk, } \text{Cov}(Y_s, Y_t) = \min(s, t)$$

Markov Property

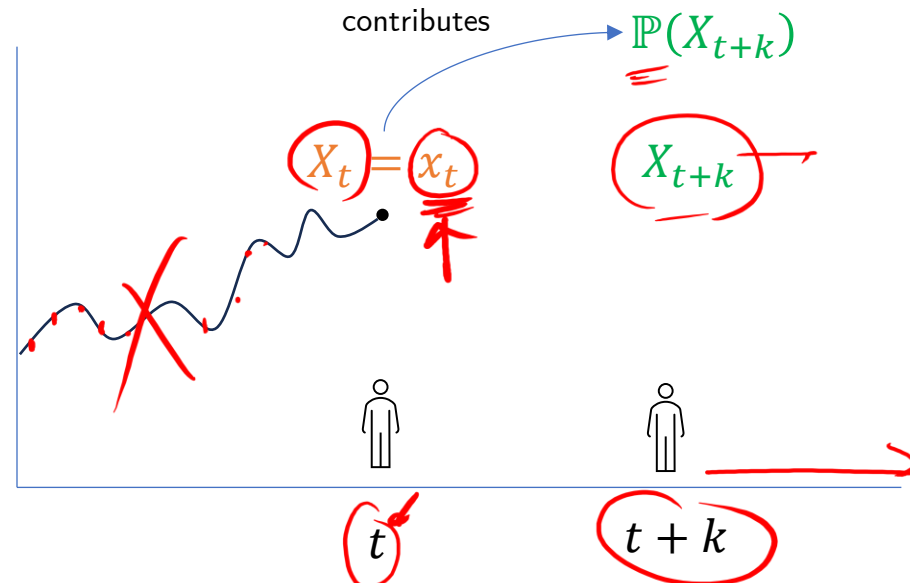
A major simplification occurs if the future distribution can be predicted only based on the present value regardless of the history of the process. This is called **Markov Property**. Markov is **memoryless**. Stated formally

$$\mathbb{P}(X_{t+k} \mid \overbrace{X_0 = x_0, X_1 = x_1, \dots, X_t = x_t}^{\text{(History)}}) = \mathbb{P}(X_{t+k} \mid \overbrace{X_t = x_t}^{\text{(Last Obs.)}})$$

(Note: The original image has red scribbles over the left side of the equation, indicating that the history part is irrelevant for the Markov property.)

Only the last observation contributes

$$\underline{y_{t+1}} = a \underline{y_t} + \epsilon_t$$



Applications of Markov

The applications of Markov is immense. It's the foundational assumption in insurance and financial engineering. All stochastic processes used to model prices are predominantly Markov. To highlight a couple of use cases

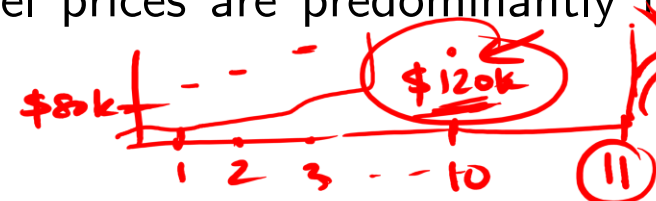
S&P 1 year (Credit Rating transition probability matrix)

	T + 1							
	AAA	AA	A	BBB	BB	B	CCC	D
AAA	92.29	6.96	0.54	0.14	0.06	0	0	0
AA	0.64	90.75	7.81	0.61	0.07	0.09	0.02	0.01
A	0.05	2.09	91.38	5.77	0.45	0.17	0.03	0.051
BBB	0.03	0.2	4.23	89.33	4.74	0.86	0.23	0.376
BB	0.03	0.08	0.39	5.68	83.1	8.12	1.14	1.464
B	0	0.08	0.26	0.36	5.44	82.33	4.87	6.663
CCC	0.1	0	0.29	0.57	1.52	10.84	52.66	34.03
D	0	0	0	0	0	0	0	100

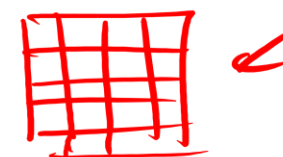
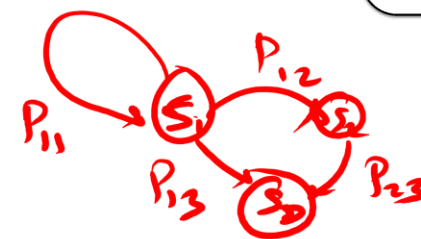
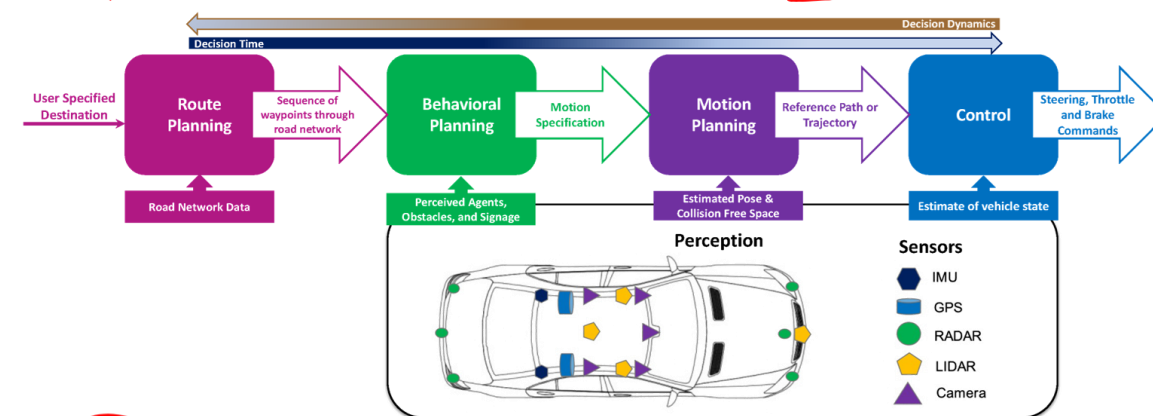
Sidebar:

A DTDS markov process is called 'Markov Chain'

A CTDS markov process is called 'Markov Jump'



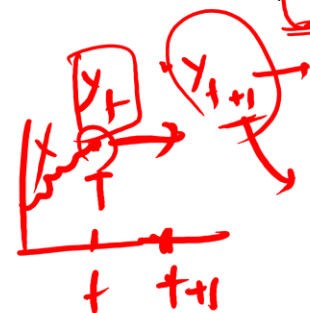
Autonomous Driving (a large scale MDP)



Independent Increments and Markov

An important result is that **independent increments imply markov** property. The reverse is not true.

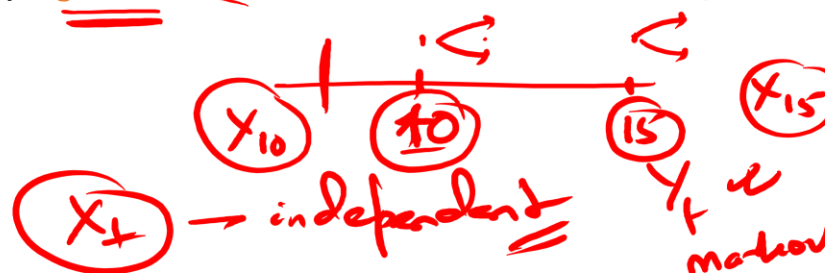
Proof : $\mathbb{P}(X_t | X_0 = x_0, X_1 = x_1, \dots, X_s = x)$



$$= \mathbb{P}(X_t - X_s + x | X_0 = x_0, X_1 = x_1, \dots, X_s = x)$$

$$= \mathbb{P}(X_t - X_s + x | X_s = x) \leftarrow (X_t - X_s) \text{ Independent increments}$$

$$= \mathbb{P}(X_t | X_s = x)$$



$$X_t = X_{t-1} + I_t$$

Question –

- 1) Prove that Random Walk has markov property.
- 2) Consider a discrete time process on integers defined as $X_t = X_{t-1} + I_t$ where I_t takes a random value of $+1$ with prob $= e^{-|X_{t-1}|}$ and value -1 with prob $= 1 - e^{-|X_{t-1}|}$. Does this process have independent increments ? Is the process Markov ?

White Noise

- ▶ White noise is a stochastic process that consists of a set of **independent and identically distributed random variables**. The random variables can be either discrete or continuous and the time set can be either discrete or continuous.
- ▶ Commonly, white noise is also considered to be **zero mean**. The **covariance of white noise is zero** for any two time points (s, t) where $s \neq t$
- ▶ The white noise process trivially satisfies strict stationarity.

In Random Walk model, the increments are white noise process.

$$X_t = \begin{cases} 1, & p \\ -1, & 1-p \end{cases}$$

$$Y_t = Y_{t-1} + X_t$$

(White Noise increment)
 $X=1$ $Y_1=1$

$$X \sim N(0, 1)$$

$p=0.5$

$X_{10}, X_{11}, \dots, X_{1000}, X_{1001}$

$X_{20}, X_{21}, \dots, X_{1000}, X_{1001}$

A general random walk is defined as a stochastic process Y_t such that $Y_n = \sum_{i=1}^n X_i$ with the condition of $Y_0 = 0$. The process has stationary independent increments and therefore Markov. The process itself is not even weakly stationary.

Sample Paths

- ▶ A joint realization of the random variables X_t for all $t \in J$ is called a **sample path** of the process.
It's a function from $J \rightarrow S$

GRW

$$Y_t = Y_{t-1} + \epsilon_t$$

$$\epsilon_t \sim \text{i.i.d.}(0)$$

(H, T)

$\omega = H, T, \dots$

HHH
HHT
HTH
HTT



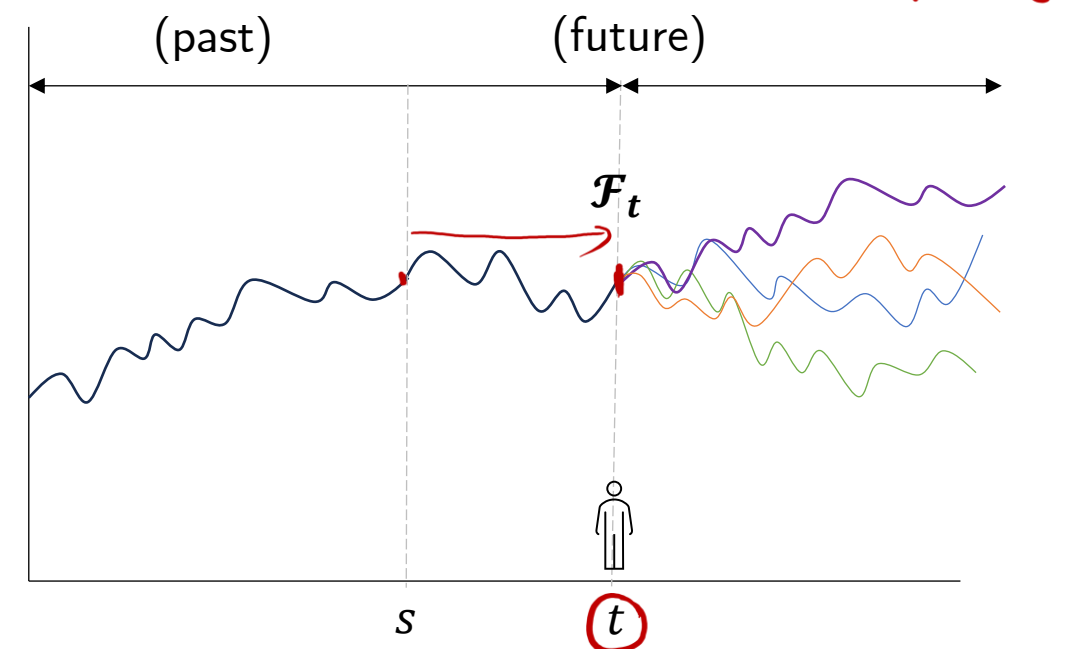
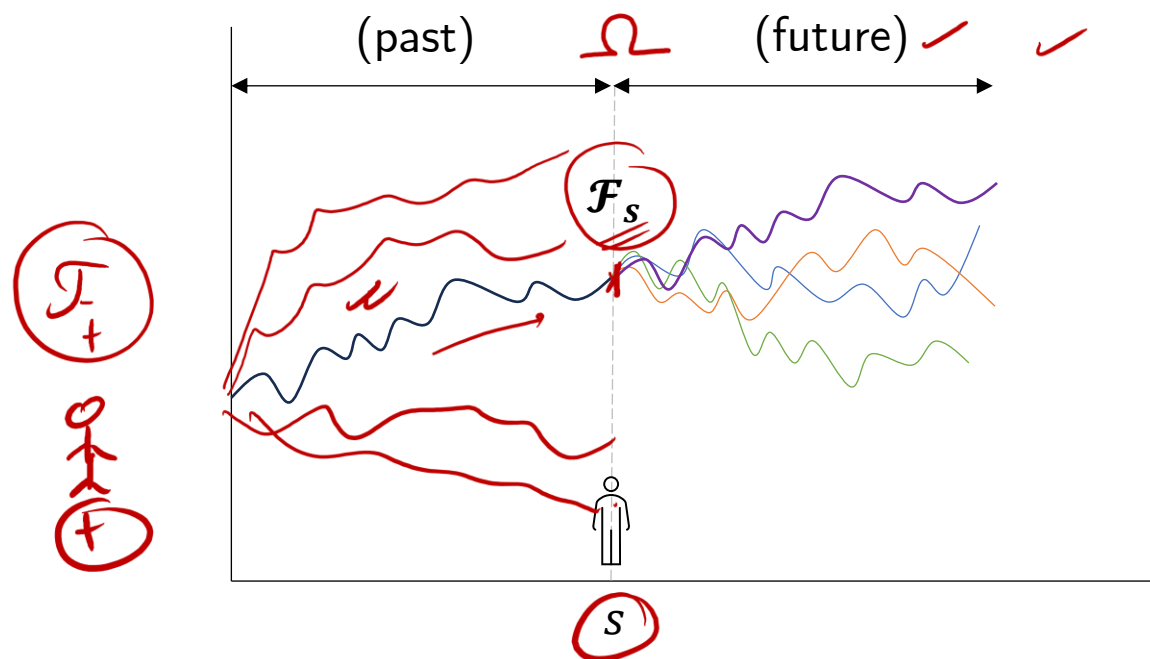
(path 1)

ω

(path 2)

Filtration (I)

- ▶ Each sample path represents one outcome (ω).
- ▶ The entire set of all possible sample paths is the sample space : Ω ✓
- ▶ Collection of all sub-sets of Ω ^(events) is called Filtration $\mathcal{F}$ which is the record of information
- ▶ Filtration is a sequence of σ -algebra such that if $s < t$, then $\mathcal{F}_s \subset \mathcal{F}_t$ ✓



Filtration (II)

Filtration gives a notational simplicity to condition stochastic processes based on information available.
For example, Markov property can be defined as

$$\mathbb{P}(X_t \leq x | \mathcal{F}_s) = \mathbb{P}(X_t \leq x | X_s)$$

for all $t \geq s \geq 0$

Handwritten notes: $F(x) = \mathbb{P}(X \leq x)$ (cdf), $\mathbb{P}(X_t \leq x | \mathcal{F}_s)$ (cdf), $\mathbb{P}(X_t \leq x | X_s)$ (last value), $\mathcal{F}_s$ (history).

A stochastic process X_t is called adapted to filtration $\mathcal{F}_t$ if it's known at time t . We also say it's $\mathcal{F}_t$ measurable. For example, X_0 is $\mathcal{F}_0$ measurable, but X_1 is not $\mathcal{F}_0$ measurable.

$$t \rightarrow X_t$$

Counting Process

Counting process is a fundamental stochastic process. A counting process is a stochastic process $\{N(t), t \geq 0\}$ with values that are non-negative, integer and non-decreasing.

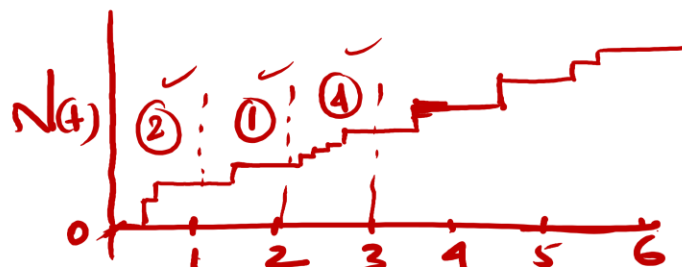
- 1) $N(t) \geq 0$ ✓
- 2) $N(t)$ is an integer ✓
- 3) If $s \leq t$, then $N(s) \leq N(t)$ ✓

Counting process deals with number of occurrences of something over time, e.g., number of road accidents over time.

Poisson Process

A **Poisson process** with rate λ is a continuous-time integer valued process N_t , $t \geq 0$ with the following properties

- 1) $N_0 = 0$
- 2) N_t has independent increments
- 3) N_t has Poisson distributed stationary increments



$\lambda = 2 \text{ occ/yr.}$

$P(X = 5) \text{ in } (2 \text{ yrs})$

$\lambda(2) = 4$

$\frac{e^{-4} \cdot 4^5}{5!}$

$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

$\mathbb{P}(N_t - N_s = n) = \frac{(\lambda(t-s))^n e^{-\lambda(t-s)}}{n!}$

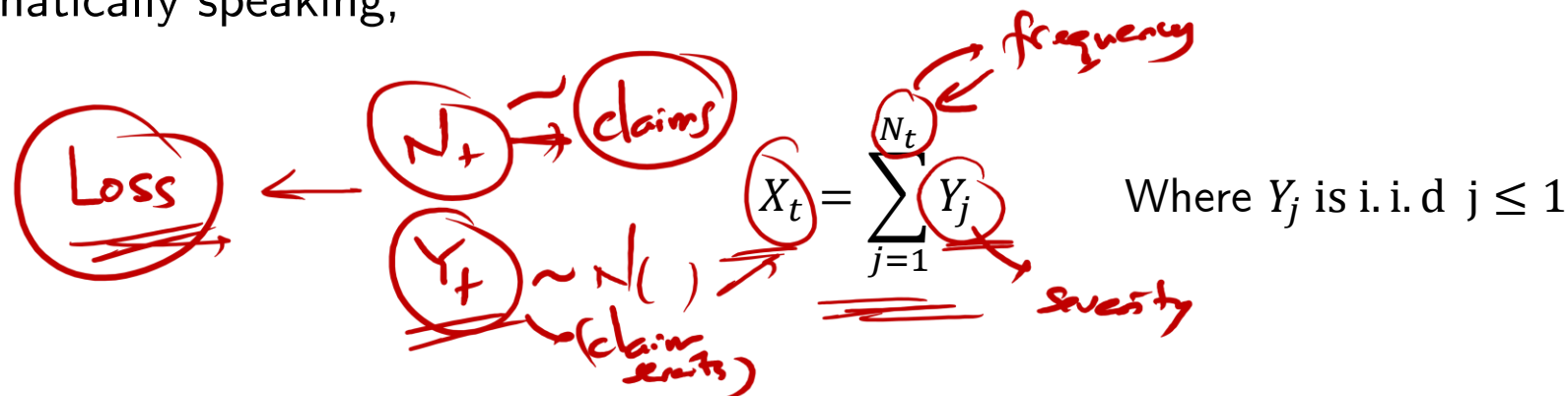
This is markov jump process with state space $S = \{0,1,2, \dots\}$. It's non-stationary as both mean and variance scale linearly with time.

$\mathbb{E}(N_t) = \lambda t$
 $\mathbb{V}(N_t) = \lambda t$
 $P(N_t = n) = \frac{e^{-(\lambda t)} \cdot (\lambda t)^n}{n!}$

Question - Calculate the covariance between the values $X(t)$ and $X(t+s)$ taken by a Poisson process $X(t)$ with constant rate λ at the two times t and $t+s$, where $s > 0$

Compound Poisson Process

A **Compound Poisson process** represents sum of random count (N_t) of i.i.d random variables. Mathematically speaking,



Y_j may be discrete or continuous. This process is used when one needs to combine **severity** and **frequency**.

For example, it's heavily used in insurance, number of claims and claim amount are random.

Also, in financial engineering where stock price is modeled with jumps (frequency of jumps and jump size)

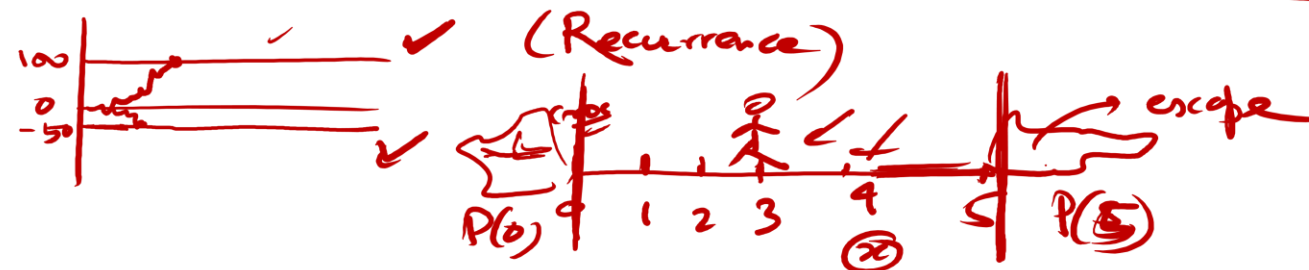
Challenging Questions

1. Consider a simple random walk process Y_t , where the increment has up-move probability of p . Show that

$$\mathbb{P}(Y_n = y) = \binom{n+y}{\frac{n+y}{2}} p^{\frac{n+y}{2}} (1-p)^{\frac{n-y}{2}}$$

2. Consider a coin toss game where you win \$1 if the coin shows H and lose \$1 if the coin shows T. The game stops if you win \$100 or lose \$100. What is the probability that the game stops with winning \$100 ? What is the probability that the game stops with losing \$100 ?

3. Consider a slight change to the above problem. The game stops after you win \$100 or lose \$50. What are the probabilities ?



Reading : Gambler's Ruin